

DR. H. S. WALL'S THEORY OF EQUATIONS COURSE

SAM CRESWELL $\ll$ H. S. WALL

THEORY OF EQUATIONS

by

H. S. Wall

Professor of Mathematics

The University of Texas

¹the relation $X \ll Y$ indicates that these notes were originally authored by Y and subsequently prepared for submission by X without significant modification and with appropriate permissions. While Y is credited with the genesis of these notes, X takes full responsibility for the current version.

Introduction by Sam Creswell

The Mathematical Association of America's recent re-publication of Prof. H. S. Wall's **Creative Mathematics** emphasizes Prof. Wall's contributions to Inquiry Based Learning.

After a career at the Institute of Advanced Study, Northwestern University, and the Illinois Institute of Technology, Prof. Wall became a colleague of Prof. R. L. Moore at the University of Texas and began teaching exclusively under the 'Moore Method' or 'Texas Method'.

Prof. Wall developed **Theory of Equations** for use in a sophomore-level course of that name at the University of Texas. The 'text' was available at an Austin bookstore for \$2.75, which was bargain even in the 1960s!

The course did not develop the notion of a limit and was independent of his calculus course. Even so, the approach is similar to that of **Creative Mathematics**. I do not know which text preceded the other. It is certain that **Creative Mathematics** and **Theory of Equations** flow from the same head-waters.

As I read through Chapter IV of these notes, I saw that Dr. Wall had not explained the derivation of the graph which appears herein as Figure 5. As I knew that Dr. Wall always offered clear expositions, relegating questions to theorems, lemmas and exercises, I took the liberty of adding a labeled remark and some graphs to the narrative in Chapter IV. Excepting typos in the original text, the balance is a re-typing¹ of the original. If a flaw has escaped my attention, please report it to me.

Dr. Wall was committed to teaching mathematics to all, not just those who were already destined for success in mathematics. This course is a rare attempt by a mathematician of the highest order to introduce students with no background in mathematics to genuine mathematics, starting with concepts with which they are familiar, while using as little jargon as possible. As such, I feel strongly that these notes need to be made accessible, rather than simply archived.

The material in these notes gets harder as the course progresses. The one-semester course that I took in the 1960s at the University of Texas did not get to Chapter IV. Even though I have never had an opportunity to teach from these notes, Chapters I and II might consume a semester for an average class. As Chapter II introduces inner-products and norms (called absolute values), a background is provided for subsequent study. As many students entering a sophomore-level course have no previous experience producing a proof, special attention might be devoted to Chapter I.

¹Proof-reading by Jermain Gonzalez, John Spellmann, John Neuberger, John Bales, Walker Hunt and Sam Young caught many of my typing errors. The editor's helpful comments also improved the article.

The use of Chapter IV as part of a one-semester course for sophomores seems problematic, at best. Indeed, the theorems are hard to even state. Chapters I and II, with some of Chapter III, may fully challenge many classes, as was the case with the class that I took at Texas.

The H. S. Wall papers are preserved in the Archives of American Mathematics, Center for American History, University of Texas at Austin. A photocopy of the original mimeographed version of **Theory of Equations** is referenced as H. S. Wall papers, 1902, 1929-1974, 2006 in the Archives of American Mathematics. The AAM's assistance is gratefully acknowledged.

These notes are submitted to the **Journal for Inquiry Based Learning in Mathematics** so another generation of students might profit from their use.

Sam Creswell
Mathematics Department
San Antonio College
San Antonio, Texas 78212-4299
screswell@alamo.edu

Table of Contents

Chapter I	The Number System	page 4
Chapter II	Some Plane geometry	page 12
Chapter III	Polynomials	page 21
Chapter IV	Location of Roots	page 29

CHAPTER I

THE NUMBER SYSTEM

Mathematics is based upon the number system consisting of :

- (a) a set of undefined elements called numbers having as a subset the set of counting numbers,
- (b) the operation of addition, denoted by $+$, such that, if x is a number and y is a number, then $x + y$, read x plus y , is a number called the sum of x and y ,
- (c) the operation of multiplication, denoted by $\cdot$, such that, if x is a number and y is a number, then $x \cdot y$, read x times y , is a number called the product of x and y ,
- (d) the order relation, denoted by $<$, and read is less than, and
- (e) a set of axioms, i.e., statements which are taken for granted.

Definitions : The symbol $=$ means is, $\neq$ means is not, $\nless$ means not less than; $x > y$, read x is greater than y , means $y < x$, $x \leq y$, read x is less than or equal to y , means $y \nless x$; and $x \geq y$, read x is greater than or equal to y , means $x \nless y$.

1. Axioms I - IV and theorems based thereon. We now state Axioms I-IV and some of their consequences.

AXIOM I

If each of x , y , and z is a number, $x + (y + z) = (x + y) + z$.

AXIOM II

0 is a number such that, if x is a number, $0 + x = x$.

AXIOM III

If x is a number, $-x$ is a number such that $x + (-x) = 0$.

AXIOM IV

If x and y are numbers, $x + y = y + x$.

Axiom I states that, if each of x , y , and z is one of the undefined elements called numbers, then x plus the number $(y + z)$ is the number $(x + y)$ plus z ; Axiom II asserts the existence of a certain number, denoted by 0, such that $0 + x$ is x for every number x ; Axiom III states that there is paired with each number x a number $-x$ such that $x + (-x)$ is 0; and Axiom IV states that $x + y$ is $y + x$.

The theorems, which are logical consequences of the axioms, furnish answers to such questions as :

- (a) Is there any other number with the property of the number 0 postulated in Axiom II?

(b) If x is a number, is there any other number with the property of the number $-x$ postulated in Axiom III?

(c) If x is a number and $-x$ is the number paired with x in accordance with Axiom III, is $-(-x)$ the number x ?

Theorem 1 : If y is a number such that, for some number x , $y + x = x$, then $y = 0$.

To prove a theorem means to supply an argument showing that the statement made in the theorem is a consequence of the axioms.

We shall prove Theorem 1.

Suppose y is a number such that, for some number x , $y + x = x$. By Axiom III, there is a number $-x$ such that $x + (-x) = 0$. Since the number $y + x$ is the number x , then

$$(y + x) + (-x) = x + (-x) \quad \text{by Axiom III.}$$

Hence, by Axiom I,

$$y + [x + (-x)] = x + (-x) \quad \text{by Axiom I,}$$

or

$$y + 0 = 0 \quad \text{by Axiom III,}$$

By Axiom IV, $y + 0$ is $0 + y$, so that

$$0 + y = 0 \quad \text{by Axiom IV.}$$

Therefore,

$$y = 0 \quad \text{by Axiom II.}$$

The number 0 such that, for each number x , $0 + x = x$, is called zero.

The problem of proving the subsequent theorems is left to the reader.

Theorem 2 : If x is a number and y is a number such that $x + y = 0$, then $y = -x$.

If x is a number, the number $-x$ such that $x + (-x) = 0$ is called the negative of x .

Theorem 3 : If x is a number, $-(-x) = x$.

Theorem 4 : If y is a number and z is a number, there is only one number x such that $y + x = z$.

If y is a number and z is a number, the number x such that $y + x = z$ is denoted by $z - y$ and called the difference z minus y . Of course, $z - y = z + (-y)$.

2. Axioms V – X and theorems based on Axioms I - X. Axioms V - VIII should be compared with axioms I - IV.

AXIOM V

If each of x , y , and z is a number, $x \cdot (y \cdot z) = (x \cdot y) \cdot z$.

AXIOM VI

1 is a number such that, if x is a number, $1 \cdot x = x$.

AXIOM VII

If x is a number such that $x \neq 0$, then $\frac{1}{x}$ is a number such that $x \cdot \frac{1}{x} = 1$.

AXIOM VIII

If x and y are numbers, $x \cdot y = y \cdot x$.

Theorem 5 : If y is a number such that, for some number x distinct from 0, $y \cdot x = x$, then $y = 1$.

The number 1 such that for each number x , $1 \cdot x = x$ is called one.

AXIOM IX

$$0 \neq 1.$$

AXIOM X

If each of x , y , and z is a number, $x \cdot (y + z) = x \cdot y + x \cdot z$.

Theorem 6 : If x is a number, $0 \cdot x = 0$.

Theorem 7 : If each of x and y is a number and $x \cdot y = 0$, then $x = 0$ or $y = 0$.

Theorem 8 : If x is a number and y is a number such that $x \cdot y = 1$, then

$$y = \frac{1}{x}.$$

If x is a number distinct from 0, the number $\frac{1}{x}$ such that $x \cdot \frac{1}{x} = 1$ is called the reciprocal of x .

Theorem 9 : If x is a number, $-1 \cdot x = -x$.

Theorem 10 : $-0 = 0$.

Theorem 11 : If each of x and y is a number, $-x \cdot y = -(x \cdot y)$.

Theorem 12 : If each of x and y is a number, $(-x) \cdot (-y) = x \cdot y$.

Theorem 13 : If each of x and y is a number, $-(x + y) = -x - y$.

Theorem 14 : If y is a number distinct from 0 and z is a number, there exists only one number x such that $y \cdot x = z$.

If y is a number distinct from 0 and z is a number, the number x such that $y \cdot x = z$ is denoted by

$$\frac{z}{y}$$

and called the quotient of z over y . Of course, $\frac{z}{y} = z \cdot \frac{1}{y}$.

Theorem 15 : If z is a number, $\frac{z}{1} = z$.

Theorem 16 : If each of x, y, u , and v is a number, $u \neq 0$ and $v \neq 0$, then

$$\frac{x}{u} \cdot \frac{y}{v} = \frac{x \cdot y}{u \cdot v}.$$

Theorem 17 : If x is a number and y is a number distinct from 0, then

$$-\frac{x}{y} = \frac{-x}{y} = \frac{x}{-y}.$$

Theorem 18 : If each of a and b is a number,

$$(a + b) \cdot (a + b) = a \cdot a + a \cdot b + b \cdot a + b \cdot b$$

and

$$(a + b) \cdot (a - b) = a \cdot a - b \cdot b.$$

Theorem 19 : If each of a, b, c , and d is a number, $c \neq 0$, and $d \neq 0$,

$$\frac{a}{c} + \frac{b}{d} = \frac{a \cdot d + c \cdot b}{c \cdot d}.$$

3. Some Examples. Inasmuch as Theorems 1 - 19 can be proved on the basis of Axioms I through X, it follows that Theorems 1 through 19 form a part of any system satisfying Axioms I through X.

Exercise 1. The system consists of two elements, 0 and 1, with addition and multiplication defined by the following tables satisfies Axioms I through X:

+	0	1
0	0	1
1	1	0

and

·	0	1
0	0	0
1	0	1

Exercise 2 : A point is an ordered pair (x, y) having a first number x , called the abscissa, and a second number y , called the ordinate. The system consisting of the set of all points with addition, denoted by $\oplus$, and multiplication, denoted by $\odot$, defined below satisfies Axioms I - X:

$$(x, y) \oplus (u, v) = (x + u, y + v)$$

and

$$(x, y) \odot (u, v) = (x \cdot u - y \cdot v, x \cdot v + y \cdot u).$$

The "zero point" is $(0, 0)$, the "one point" is $(1, 0)$, the negative of point (x, y) is $(-x, -y)$ and each point distinct from $(0, 0)$ has a reciprocal.

The system consisting of the set of all points $(x, 0)$ with ordinate 0, addition, $\oplus$, and multiplication, $\odot$, being defined as above, satisfies Axioms I - X and may

be identified with the number system. Thus, if (x, y) is a point and c is a number, we shall write

$$\begin{aligned} c \oplus (x, y) &= (c + x, y) \text{ instead of } (c, 0) \oplus (x, y) = (c + x, y) \\ \text{and } c \odot (x, y) &= (c \cdot x, c \cdot y) \text{ instead of } (c, 0) \odot (x, y) = (c \cdot x, c \cdot y). \end{aligned}$$

The statement that T is a transformation from the set A to the set B means that T is a collection of one or more ordered pairs (a, b) whose first element a is in A and whose second element b is in B such that no two ordered pairs in T have the same first element, each member of A is the first element of some ordered pair in T . The second element of that ordered pair in the transformation T whose first element is a is denoted by Ta or by $T(a)$ and is read “ T of a ”.

The statement that the transformation T from a set A to set B is reversible² means that no two ordered pairs in T have the same second element. If T is a reversible transformation from the set A to the set B , then the collection of ordered pairs to which (b, a) belongs only if (a, b) belongs to T is a transformation from B to A , called the inverse of T and is denoted by T^{-1} .

The statement that the transformation T from the set of all points to a subset of the set of all points is linear means that, if P is a point, Q is a point and c a number, then

$$T(P \oplus Q) = TP \oplus TQ \text{ and } T(c \odot P) = c \odot TP.$$

Example 1 : Suppose T is the transformation such that for each point P , $TP = P$. Then, T is linear.

Example 2 : Suppose T is the transformation such that, for each point P , $TP = (0, 0)$. Then, T is linear.

Example 3 : Suppose T is the transformation such that, for each point (x, y) , $T(x, y) = (x + y, x + y)$. Then, T is linear.

Exercise 3 : Determine all the linear transformations from the set of all points to a subset thereof.

Axioms and Theorems Concerning the Order Relation.

The following axioms, XI - XVI, complete the set of axioms for the number system.

Axiom XI

If x and y are numbers, $x < y$ or $y < x$.

Axiom XII

If each of x , y , and z is a number and $x < y$, then $x + z < y + z$.

Axiom XIII

If each of x , y , and z is a number, $x < y$ and $z > 0$, then $x \cdot z < y \cdot z$.

²Invertible and one-to-one are synonyms for reversible.

Axiom XIV

If each of x , y , and z is a number, $x < y$ and $y < z$, then $x < z$.

Axiom XV

Suppose M is a number set. If there is a number which no number in M exceeds, there is a least number which no number in M exceeds. If there is a number which exceeds no number in M , there is a largest number which exceeds no number in M .

Axiom XVI

1 is a counting number and if h is a number less than 1, h is not a counting number. If k is a counting number, then $k + 1$ is a counting number; and if h is a number such that $k < h$ and $h < k + 1$, then h is not a counting number.

Theorem 20. If x is a number and y is a number, only one of the following statements is true : $x < y$, $x = y$ or $x > y$.

Theorem 21. $0 < 1$.

Theorem 22. If x is a number and $x > 0$, then $\frac{1}{x} > 0$.

Theorem 23. If x and y are numbers such that $x > 0$ and $y > 0$ and $x < y$, then

$$\frac{1}{x} > \frac{1}{y}.$$

Theorem 24. If x and y are numbers and $x > y$, there exists a number k such that $k > 0$ and $x = y + k$.

Theorem 25. If x is a number, $x < x + 1$.

Theorem 26. If each of x , y , u , and v is a number, $x < y$ and $u < v$, then $x + u < y + v$.

Theorem 27. If x and y are numbers and $x < y$, then $-y < -x$.

The statement that the number x is between the number y and the number z means that $y < x$ and $x < z$ or $z < x$ and $x < y$. The statement that the number x is positive means that $x > 0$ and the statement that x is negative means that $x < 0$.

Theorem 28. Suppose each of p and q is a positive number. Then,

$$p + q > p \text{ and } p + q > q;$$

$$\frac{p \cdot q}{p + q} < p \text{ and } \frac{p \cdot q}{p + q} < q;$$

and, if $p \neq q$,

$\frac{p+q}{1+1}$ is between p and q .

5. Counting Numbers. Axiom XVI specifies the counting numbers.

Theorem 29. If p is a number, there is at most one counting number between p and $p + 1$.

Theorem 30. If M is a set each element of which is a counting number, there is a number in M which exceeds no number in M .

Theorem 31. If h is a number greater than 1 which is not a counting number, there exists a counting number k such that h is between k and $k + 1$.

Theorem 32. If k is a counting number greater than 1, then $k - 1$ is a counting number.

Theorem 33. If G is a collection of counting numbers, 1 is in G , and for each x in G , $x + 1$ is in G , then every counting number is in G .

Theorem 34. If m is a counting number and n is a counting number, then $n + m$ is a counting number and $m \cdot n$ is a counting number.

Theorem 35. If δ is a positive number and k is a positive number, then there exists a counting number n such that $n \cdot \delta > k$.

6. Rational Numbers. Suppose that x is a number. The statement that x is a positive integer means x is a counting number; the statement that x is a negative integer means that x is the negative of a counting number; the statement that x is an integer means that x is a positive integer, x is 0, or x is a negative integer; and the statement that x is a rational number means that there exist an integer m and an integer n such that $x = \frac{m}{n}$.

Theorem 36. If x and y are numbers, there is a rational number between x and y .

Theorem 37. The statements obtained from Axioms I - XIV and XVI by the replacement of "number" by "rational number" are true.

Theorem 38. There exists a positive number h such that $h \cdot h = 1 + 1$.

$1 + 1 = 2$, $2 + 1 = 3$, $3 + 1 = 4$, ... ; if x is a number, $x \cdot x = x^2$, $x^2 \cdot x = x^3$, $x^3 \cdot x = x^4$, ..., read " x squared", " x cubed", " x to the fourth power", respectively.

If a is a number and b is a number, ab means $a \cdot b$.

Theorem 39. If a is a positive number, there exists only one positive number h such that

$$h^2 = a.$$

If a is a positive number, the positive number h such that $h^2 = a$ is called the positive square root of a and is denoted by $\sqrt{a}$.

Theorem 40. The positive square root of 2 is not a rational number.

Theorem 41. If a is a number and b is a number, $a^2 + 2ab + b^2 > 0$, unless $a = -b$.

Theorem 42. If each of a , b , c and x is a number and $a \neq 0$, then

$$ax^2 + bx + c = a \left(\left(x + \frac{b}{2a} \right)^2 - \frac{b^2 - 4ac}{4a^2} \right).$$

CHAPTER II

SOME PLANE GEOMETRY

The point (x, y) may be pictured as a dot on a writing surface at a certain distance, measured in terms of any convenient unit, horizontally and vertically from a chosen place 0. If $x > 0$ and $y > 0$, the dot is a distance x to the right of 0 and a y upward; if $x < 0$ and $y > 0$, the dot is $-x$ to the left of 0 and $-y$ downward; if $x > 0$ and $y < 0$, the dot is x to the right of 0 and $-y$ downward; the point $(x, 0)$ is pictured x to the right of 0, at 0 if $x = 0$ and $-x$ to the left of 0 if $x < 0$; and $-y$ below 0 if $y < 0$. To each point (x, y) there corresponds a dot on the writing surface and to each dot on the writing surface there corresponds a point (x, y) .

This scheme for picturing point sets enables us to unite geometry and the number system.

1. Matrices. The square array

$$\begin{bmatrix} a, & b \\ c, & d \end{bmatrix}$$

in which each of a, b, c , and d is a number, is called a 2×2 matrix. We shall define addition and multiplication among 2×2 matrices in such a way that all but two of Axioms I – X of Chapter I are satisfied.

Theorem 1. The system of all 2×2 matrices with addition, denoted by $\oplus$, and multiplication, denoted by $\odot$, defined by

$$\begin{bmatrix} a, & b \\ c, & d \end{bmatrix} \oplus \begin{bmatrix} A, & B \\ C, & D \end{bmatrix} = \begin{bmatrix} a + A, & b + B \\ c + C, & d + D \end{bmatrix}$$

and

$$\begin{bmatrix} a, & b \\ c, & d \end{bmatrix} \odot \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} aA + bC, & aB + bD \\ cA + dC, & cB + dD \end{bmatrix}$$

satisfied all but two of the Axioms I - X of Chapter I, the exceptions being Axiom VII and Axiom VIII. In addition to Axiom X, it is true that, if each of U, V, and W is a 2×2 matrix,

$$(U \oplus V) \odot W = (U \odot W) \oplus (V \odot W).$$

Definition. The product $\begin{bmatrix} a, & b \\ c, & d \end{bmatrix} (x, y)$ of a 2×2 matrix times a point is the point $(ax + by, cx + dy)$.

Note. If we momentarily identify the point (x, y) with the 2×2 matrix $\begin{bmatrix} x, & 0 \\ y, & 0 \end{bmatrix}$, the product $\begin{bmatrix} a, & b \\ c, & d \end{bmatrix} \odot \begin{bmatrix} x, & 0 \\ y, & 0 \end{bmatrix} = \begin{bmatrix} ax + by, & 0 \\ cx + dy, & 0 \end{bmatrix}$ identified

with the point $(ax + by, cx + dy)$. It follows that if each of U and V is a 2×2 matrix and P is a point, $U(VP) = (U \odot V)P$ and $(U \odot V)P = UP + VP$.

Henceforth we shall use the symbol $+$ to denote addition among numbers, among points and among matrices. Also, multiplication will be denoted by $a \cdot b$ or ab in every case.

Theorem 2. Let S denote the set to which T belongs if T is a linear transformation from the set of all points to a subset thereof and M the set of all 2×2 matrices. There exists a reversible transformation W from S to M such that (T, A) is an ordered pair in W only if, for every point P ,

$$TP = AP.$$

In accordance with this theorem we may, in effect, identify the set S with the set M . That is, we need not make a distinction between linear transformations of S and the 2×2 matrices of M . Also, if c is a number and P is a point, there is no need to distinguish between the two products $\begin{bmatrix} c, & 0 \\ 0, & c \end{bmatrix} P$ and CP inasmuch as either product is the point (cx, cy) , P being (x, y) . Finally, if C is the point (a, b) , there is no need to distinguish between the two products $\begin{bmatrix} a, & -b \\ b, & a \end{bmatrix} P$ and CP .

2. The Number Plane. The system consisting of the set of all points, with addition and multiplication among the points defined as in Exercise 2, Section 3 of Chapter I, together with a notion of distance, to be defined presently, is called the number plane. We recall that addition is defined by

$$(x, y) + (u, v) = (x + u, y + v)$$

and multiplication by

$$(x, y) \cdot (u, v) = (xu - yv, xv + yu).$$

If $P = (x, y)$ and $Q = (u, v)$, the inner product of P and Q , denoted by $((P, Q))$, is the number $xu + yv$.

Theorem 3. The inner product has the following properties:

- (a) if P is a point and Q is a point, then $((P, Q)) = ((Q, P))$,
- (b) if each of P , Q and R is a point, then $((P + Q, R)) = ((P, R)) + ((Q, R))$,
- (c) if each of P and Q is a point and c is a number, then $((cP, Q)) = c((P, Q))$, and
- (d) If P is a point, then $((P, P)) > 0$, unless $P = (0, 0)$.

Definition. If P is a point, the nonnegative number whose square is $((P, P))$ is denoted by $|P|$ and called the absolute value of P . Thus, if $P = (x, y)$:

$$|P| = \sqrt{((P, P))} = \sqrt{x^2 + y^2}.$$

Theorem 4. If each of P and Q is a point, then

$$((P, Q))^2 \leq |P|^2 \cdot |Q|^2$$

and, if each of P , Q , and R is a point, then

$$|P - Q| + |Q - R| \geq |P - R|.$$

Definition. If each of P and Q is a point, $|P - Q|$ is called the distance between P and Q .

Note that $|P - Q| = |Q - P| > 0$, unless $P = Q$, if c is a number, $|cP| - |cQ| = |c| \cdot |P - Q|$ and distance has the triangle property :
 $|P - Q| + |Q - R| \geq |P - R|.$

Theorem 5. Suppose T is a linear transformation from the set of all points to a subset thereof (i.e., T is a 2×2 matrix). The following two statements are equivalent:

(1) there exists an ordered number pair a, b such that

$$T = \begin{bmatrix} a, & -b \\ b, & a \end{bmatrix} \text{ or } \begin{bmatrix} a, & b \\ b, & -a \end{bmatrix}$$

and

(2) there exists a nonnegative number c such that, if x is a point and y is a point,

$$|Tx - Ty| = c|x - y|.$$

Let G denote the set to which the 2×2 matrix T belongs only if there exists an ordered number pair a, b such that

$$T = \begin{bmatrix} a, & -b \\ b, & a \end{bmatrix}.$$

Theorem 6. The set G , with addition and multiplication as defined among matrices, satisfies Axioms I – X of Chapter I.

As noted in Section 1, $\begin{bmatrix} a, & -b \\ b, & a \end{bmatrix} (x, y) = (a, b) \cdot (x, y)$. Denote by U the 2×2 matrix $\begin{bmatrix} 1, & 0 \\ 0, & -1 \end{bmatrix}$. Then,

$$\begin{bmatrix} a, & -b \\ b, & a \end{bmatrix} U = \begin{bmatrix} a, & b \\ b, & -a \end{bmatrix} \text{ and } U(x, y) = (x, -y).$$

The 2×2 matrix T has the property specified in statement (2) of Theorem 5 only if T is in G or there exists an element S in G such that $T = SU$.

Definition. If P is a point (x, y) , $\overline{P}$ is the point $(x, -y)$ and is called the conjugate of P .

Theorem 7. If each of P and Q is a point, the following statements are true:

- (1) $P \cdot \overline{P} = |P|^2$,
- (2) $\overline{P + Q} = \overline{P} + \overline{Q}$,
- (3) $|\overline{P}| = |P|$,

- (4) $|PQ| = |P| \cdot |Q|,$
 (5) $|P + Q| \leq |P| + |Q|,$ and
 (6) $|P - Q| \geq |P| - |Q|.$

Moreover, if $P \neq 0,$

$$(1) \quad \frac{1}{P} = \frac{1}{|P|^2} \cdot \bar{P} \quad \text{and}$$

$$(2) \quad \overline{\left(\frac{Q}{P}\right)} = \frac{\bar{Q}}{\bar{P}}.$$

Exercise 1. Suppose that each of P and Q is a point. Give a geometrical interpretation of the sum $P + Q$ and of the product PQ .

Exercise 2. Find a point a such that $a \cdot a = (-1, 0)$, i.e., $a^2 = -1$.

3. Straight Lines and Circles If P and Q are points, $L(P, Q)$ denotes the point set to which the point R belongs only if there exists a number t such that $R = (1 - t)P + tQ$. If t is a number, the point $(1 - t)P + tQ$, denoted by R_t is the point of $L(P, Q)$ which corresponds to t .

Theorem 8. Suppose each of u, v , and w is a number, P and Q are points and for each number t , R_t is the point of $L(P, Q)$ which corresponds to t , i.e., $R_t = (1 - t)P + tQ$. The following statements are true:

- (1) $R_u - R_v = (u - v) \cdot (Q - P)$ so that $R_u = R_v$ only if $u = v$,
 (2) if $u < v < w$, then $|R_v - R_u| + |R_w - R_v| = |R_w - R_u|$, and
 (3) if U and V are points of $L(P, Q)$, then $L(U, V) = L(P, Q)$, i.e., if R is a point of either of the sets $L(U, V)$ or $L(P, Q)$, then R is in the other.

Theorem 9. If P and Q are points and S a point not in $L(P, Q)$, there is a point T in $L(P, Q)$ which is nearer S than any other point of $L(P, Q)$. This point T is the only point X in $L(P, Q)$ such that $((P - Q, S - X)) = 0$.

Theorem 10. Suppose T is the 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (linear transformation). T is reversible only if $ad - bc \neq 0$.

If M is a point set and f a transformation from a set A which includes M to a set B , then $f\{M\}$ denotes the set to which y belongs only if there is a point x in M such that $y = f(x)$.

Theorem 11. Suppose P and Q are points and T is a reversible linear transformation from the set of all points to a subset thereof. Then, $T\{L(P, Q)\} = L(TP, TQ)$.

Definition. Suppose P and Q are points and P_1 and Q_1 are points. The statement that $L(P, Q)$ is parallel to $L(P_1, Q_1)$ means that there is a number c such that $P - Q = c(P_1 - Q_1)$.

Theorem 12. Suppose P and Q are points, P_1 and Q_1 are points and $L(P, Q)$ is parallel to $L(P_1, Q_1)$. If T is a reversible linear transformation from the set of all points to a subset thereof, then $T\{L(P, Q)\}$ is parallel to $T\{L(P_1, Q_1)\}$.

Suppose a is a point with absolute value 1 and p is a nonnegative number. There exists a point z such that $((z - pa, a)) = 0$. Let $M_{p,a}$ denote the point set to which z belongs only if $((z - pa, a)) = 0$.

Theorem 13. The point z belongs to $M_{p,a}$ only if

$$\bar{a}z + a\bar{z} - 2p = 0.$$

We denote by i the point $(0, 1)$, so that $i^2 = -1$, $\bar{i} = (0, -1)$ and $(-i)^2 = (\bar{i})^2 = -1$.

Theorem 14. The following statements are true :

- (1) the point (x, y) belongs to $M_{0,1}$ only if $x = 0$, i.e., $M_{0,1}$ is on the "Y-axis",
- (2) the point (x, y) belongs to $M_{0,i}$ only if $y = 0$, i.e., $M_{0,i}$ is on the "X-axis",
- (3) $L(0, ia) = M_{0,a}$,
- (4) $L(pa, ia + pa) = M_{p,a}$,
- (5) if P and Q are points, there exists a nonnegative number p and a point a with absolute value 1 such that $L(P, Q) = M_{p,a}$,
- (6) if z is a point and d the number
$$\frac{\bar{a}z + a\bar{z} - 2p}{2} \quad (|a| = 1, p \geq 0),$$
 then the point $z - ad$ of $M_{p,a}$ is nearer to z than any other point of $M_{p,a}$ and its distance from z is $|d|$, and
- (7) if $p > 0$, z is in $M_{p,a}$ only if $|z| = |z - 2pa|$; and, if $p = 0$, z is in $M_{p,a}$ only if $|z - a| = |z + a|$.

The point set $L(P, Q)$ is called the straight line determined by P and Q ; the point set $M_{p,a}$ is the straight line with equation $\bar{a}z + a\bar{z} - 2p = 0$.

The statement that the straight line determined by the points P and Q is perpendicular to the straight line determined by the points P_1 and Q_1 means that the inner product $((P - Q, P_1 - Q_1)) = 0$.

Theorem 15. Suppose P , Q , and R are points. The following two statements are equivalent :

- (1) $L(P, Q)$ is perpendicular to $L(P, R)$, and
- (2) $|Q - R|^2 = |Q - P|^2 + |P - R|^2$.

Since the straight line $M_{p,a}$ ($|a| = 1, p \geq 0$) is the straight line determined by the points pa and $ai + pa$, we see that straight line $M_{p,a}$ is perpendicular to the straight line $M_{q,b}$ ($|b| = 1, q \geq 0$) only if $((a, b)) = 0$.

The line segment with end-points P and Q is the point set to which z belongs if there exists a number t between 0 and 1 such that $z = (1 - t)P + tQ$. Note that the line segment does not contain its end-points.

The statement that $C_{c,r}$ is the circle with center c and radius r means c is a point, r a positive number, and that $C_{c,r}$ is the point set to which z belongs only if $|z - c| = r$. The interior of $C_{c,r}$, denoted by $C_{c,r}^-$, is the set to which z belongs only if $|z - c| < r$; the exterior, denoted by $C_{c,r}^+$, is the set to which z belongs only if $|z - c| > r$.

Theorem 16. If P and Q are points of $C_{c,r}^-$, then the line segment with end-points P and Q is a subset of $C_{c,r}^-$.

Theorem 17. If each of A and C is a number and B a point, there is a point Z such that

$$AZ\bar{Z} + \bar{B} + BZ + C = 0$$

only in case

- (1) $A = B = C = 0$, or
- (2) one of A, B , or C is distinct from 0 and
 - (a) $A = 0$ and $B \neq 0$, or
 - (b) $A \neq 0$ and $B\bar{B} - AC \geq 0$.

If $M_{p,a}$ is a straight line ($|a| = 1, p \geq 0$), $M_{p,a}^-$ denotes the point set to which z belongs only if $\bar{a}z + a\bar{z} - 2p < 0$, and $M_{p,a}^+$ denotes the point set to which z belongs only if $\bar{a}z + a\bar{z} - 2p > 0$. The point sets $M_{p,a}^-$ and $M_{p,a}^+$ are the half-planes bounded by $M_{p,a}$.

Theorem 18. If P and Q are points belonging to a half-plane, the line segment with end-points P and Q is a subset of the half-plane.

Theorem 19. If P is a point in one of the half-planes bounded by $M_{p,a}$ and Q a point in other, there is only one point of the segment with end-points P and Q which belongs to $M_{p,a}$.

Theorem 20. $C_{c,r}^-$ is a subset of $M_{p,a}^+$ only if

$$\frac{\bar{a}c + a\bar{c} - 2p}{2} \geq r.$$

4. Linear Fractional Transformations. The linear fractional transformation is a generalization of the linear transformation T for which there is a point C such, for every point x , $Tx = Cx$.

The extended number plane is the number plane together with one "ideal point", denoted by ∞ . The ideal point is not an ordered number pair.

Theorem 21. Suppose each of a, b, c and d is a point and $ad - bc \neq 0$. There is no point z such that $az + b = 0$ and $cz + d = 0$.

Note. We use the word "point" as usual, to denote an ordered number pair; ∞ will always be referred to as the "ideal point".

Definition. The statement that f is a linear fractional transformation means that f is a transformation from the extended number plane to a subset thereof and there exists an ordered-point, a quadruple, (a, b, c, d) such that $ad - bc \neq 0$ and (z, w) is an ordered pair in f only if z is a point or the ideal point ∞ and

$$w = \begin{cases} \frac{az + b}{cz + d}, & \text{if } z \text{ is a point and } cz + d \neq 0; \\ \infty, & \text{if } z \text{ is a point and } cz + d = 0 \text{ or } z = \infty \text{ and } c = 0; \\ \frac{a}{c}, & \text{if } z = \infty \text{ and } c \neq 0. \end{cases}$$

Theorem 22. If f is a linear fractional transformation, then f is a reversible transformation from the extended number plane to the extended number plane.

If the linear transformation f is given by

$$f(z) = \frac{az + b}{cz + d},$$

then the inverse f^{-1} is given by

$$f^{-1} = \frac{dz - b}{-cz + a}$$

and f^{-1} is a linear fractional transformation.

We adopt the convention that the ideal point ∞ belongs to every straight line.

Exercise 1. If f is a linear fractional transformation given by

$$f(z) = \frac{z}{z + 1}$$

and M is the straight line with equation $z + \bar{z} = 0$, find $f\{M\}$.

Theorem 23. If f is a linear fractional transformation and M is a circle then, $f\{M\}$ is a circle or a straight line.

Theorem 24. Suppose H is the straight line with equation $\bar{a}z + a\bar{z} - 2p = 0$ ($\|a\| = 1, p \geq 0$) and K is the straight line with equation $\bar{a}z + a\bar{z} - 2q = 0$ ($\|a\| = 1, q \geq 0$). If A is a point distinct from 0 and B is a point and f the linear fractional transformation defined by

$$f(z) = \frac{-A^2}{B - z},$$

then

- (1) $f\{K^-\}$ is the interior of a circle only if B is in K^+ , and
- (2) $f\{K^-\}$ is the interior of a circle and a subset of H^- only if

$$\bar{a}B + a\bar{B} - 2q \geq 0$$

and

$$\|A^2\| - \frac{1}{2}[(A\bar{a})^2 + (\bar{A}a)^2] \leq p[\bar{a}B + a\bar{B} - 2q].$$

We denote by $\text{Re}P$ and $\text{Im}P$ the abscissa and ordinate, respectively, of the point P . Inasmuch as $P + \bar{P} = 2\text{Re}P$, we see that the last inequality of Theorem 24 may be written :

$$|(A\bar{a})|^2 R_e(A\bar{a})^2 \leq 2p[R_e B\bar{a} - q].$$

Now, if z is the point (x, y) , then $z^2 = (x^2 - y^2, 2xy)$, so that $z^2 - R_e z^2 = 2y^2$. Therefore, our inequality may be written:

$$[I_m \bar{a} A]^2 \leq p[\text{Re} \bar{a} B - q]$$

Thus, $f\{K^-\}$ is the interior of a circle and a subset of K only if $\text{Re} a B > q$ and $[I_m a A]^2 \leq p[\text{Re} \bar{a} B - q]$.

The product fg of a linear fractional transformation f times the linear fractional transformation g is the transformation h defined by $h(z) = f[g(z)]$.

Theorem 25. The product fg of the linear transformation f times the linear fractional transformation g is a linear fractional transformation.

Suppose n is an integer greater than 1, each of $A_1^2, \dots, A_n^2$ a point distinct from 0, each of $B_1, B_2, \dots, B_n$ a point and f_i the linear fractional transformation defined by

$$f_i(z) = \frac{-A_i^2}{B_i - z}, i = 1, \dots, n.$$

Then, the product $f_1 \cdot f_2 \cdots f_n$ is the linear fractional transformation h , defined by

$$h(z) = \frac{-A_1^2}{B_1 + \frac{A_2^2}{B_2 + \cdots + \frac{A_n^2}{B_n - z}}}.$$

Theorem 26. Suppose H_i is a straight line with equation $\bar{a}z + a\bar{z} - 2p_i = 0$, ($|a| = 1, p_i > 0$),

$$\text{Re} \bar{a} B_i > p_i, \quad i = 1, \dots, n.$$

and

$$[I_m \bar{a} A_i]^2 \leq p_{i-1}[\text{Re} \bar{a} B_i - p_i], \quad i = 2, \dots, n.$$

Then, $h\{H_n^-\}$ is the interior of a circle and, in particular, $h(0)$ is not ∞ , i.e.,

$$\frac{-A_1^2}{B_1 + \frac{A_2^2}{B_2 + \cdots + \frac{A_n^2}{B_n}}}.$$

is a point (not the ideal point).

CHAPTER III

POLYNOMIALS

A polynomial is a transformation from the number plane to a subset of the number plane.

The statement that f is a polynomial of degree 0 means that there exists a point a distinct from 0 such that, for every point z , $f(z) = a$. Thus, a polynomial of degree 0 is a transformation from the set of all points to the set of the single point a , distinct from 0.

If n is a positive integer, the statement that f is a polynomial of degree n means there exists a point a distinct from 0 such that, for every point z ,

$$f(z) = az^n + g(z)$$

where $g(z) = 0$ for every point z or else g is a polynomial of degree less than n . Thus, if f is a polynomial of degree the nonnegative integer n , we may write

$$f(z) = \sum_{p=0}^n a_p z^{n-p},$$

where a_p is a point and $a_0 \neq 0$, $z^0 = 1$ and $\sum_{p=0}^n$ indicates summation.

Thus, $\sum_{p=0}^0 a_p z^{0-p} = a_0$, $\sum_{p=0}^1 a_p z^{1-p} = a_0 z + a_1$, $\sum_{p=0}^2 a_p z^{2-p} = a_0 z^2 + a_1 z + a_2$, etc.

1. Three fundamental processes. A polynomial of degree 1 is a linear fractional transformation f such that $f(\infty) = \infty$, and is a reversible transformation from the extended number plane to the extended number plane.

Exercise 1. The polynomial f defined by $f(z) = z^2$ is a transformation from the set of all points to the set of all points, which is not reversible.

Theorem 1. (The Substitution Process) Suppose f is a polynomial of positive degree n defined by

$$f(z) = \sum_{p=0}^n a_p z^{n-p}$$

and r is a point. There exists a polynomial g of degree $n - 1$ and a point R such that, for every point z ,

$$f(z) = (z - r)g(z) + R.$$

Moreover, if $g(z) = \sum_{p=0}^{n-1} b_p z^{n-1-p}$, the n -tuple $b_0, b_1, \dots, b_{n-1}$ and R may be computed by means of the process indicated by the following table:

$a_0,$	$a_1,$	$a_2,$	$\dots$	$a_{n-1},$	a_n	r
	$rb_0,$	$rb_1,$	$\dots$	$rb_{n-2},$	rb_{n-1}	

$b_0 = a_0, b_1 = a_1 + rb_0, b_2 = a_2 + rb_1, \dots, b_{n-1} = a_{n-1} + rb_{n-2}, R = a_n + rb_{n-1}$
 Note. If $n = 1$, then $b_0 = a_0$ and $R = a_1 + rb_0$; if $n = 2$, then $b_0 = a_0, b_1 = a_1 + rb_0$ and $R = a_2 + rb_1$.

Corollary. If f is a polynomial of positive degree, r is a point, and R the point given by the preceding table, then $R = f(r)$. If, in particular, r is a point such that $f(r) = 0$, then, for every point z

$$f(z) = (z - r)g(z)$$

where g is a polynomial of degree $n - 1$.

The statement that r is a root of the polynomial f means r is a point such that $f(r) = 0$.

Exercise 2. Show that a polynomial of positive degree n has not more than n roots.

Exercise 3. Show that, if r is a root of the polynomial f of degree n , there is a positive integer m not greater than n and a polynomial h of degree $n - m$ such that for every point z ,

$$f(z) = (z - r)^m h(z) \text{ and } h(r) \neq 0.$$

The positive integer m of this exercise is called the multiplicity of the root r of f . A root of multiplicity 1 is called a simple root of f .

Fundamental Theorem of Algebra.

Each polynomial of positive degree has a root. We recommend that the reader take this theorem for granted for the present.

If M is a point set, the statement that q is the multiplicity sum in M for the polynomial f means that

- (1) if f has no root in M , then $q = 0$
- (2) if f has just one root in M and this root is of multiplicity m , then $q = m$
- (3) if f has just two roots in M , one of multiplicity m_1 and the other of multiplicity m_2 , then $q = m_1 + m_2$
- (4) if f has just k roots in M , of multiplicities $m_1, m_2, \dots, m_k$, then $q = m_1 + m_2 + \dots + m_k$.

Exercise 4. Show, on the basis of the Fundamental Theorem of Algebra, that the multiplicity sum in the set of all points for the polynomial f is the degree of f .

We call the process indicated by the table for Theorem 1 the substitution process. The reason for this is that the point R is $f(r)$, the result of "substitution" of r for z in $f(z)$.

Example. Suppose $f(z) = z^3 - 3z^2 + 2z - 7$. In this case, if z is a number (i. e., a point with ordinate 0), then $f(z)$ is a number. Thus f may be regarded as a transformation of the set of all numbers into a subset thereof. A convenient way to study this "contraction of f " to the set of numbers is to draw the graph of the

points (x, y) to which x is a number and $y = x^3 - 3x^2 + 2x - 7$. The computation of the ordinates y for various abscissas x may be arranged as follows.

1	-3	2	-7	x	y
	1	-2	0	1	
1	-2	0	-7	1	-7

which means that if (x, y) is a point of the graph and $x = 1$, then $y = -7$. Using this process, Professor Wall computed the following table of points of the graph.

x	y
0	-7
1	-7
2	-7
0.5	-6.1
1.5	-7.4
3	-1
4	17
-1	-13
-2	-67
5	53

from which he sketched a graph of f . Note that $f(3) < 0$ and $f(4) > 0$.

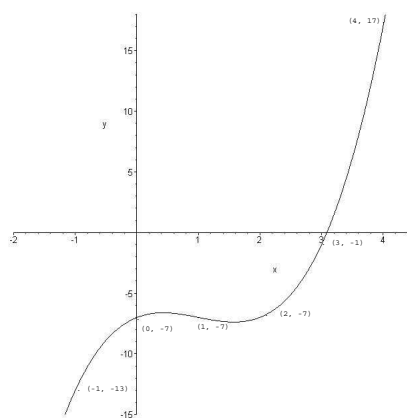


FIGURE 1. Graph of $y = -x^3 - 3x^2 + 2x - 7$

Exercise 5. Suppose f is a polynomial of positive degree n and $f(z) = \sum_{p=0}^n a_p z^{n-p}$, where each coefficient a_p is a number. The following statements are true :

- (1) if c is a number, there exists a polynomial g of degree $n - 1$, each coefficient of which is a number, such that $f(x) = (c - x)g(x) + f(c)$ for every number x ;
- (2) if c is a number such that $f(c) \neq 0$, there exists two numbers a and b with c between them such that $f(x) > 0$ for every number x between a and b or else $f(x) < 0$ for every number x between a and b ;

- (3) if a and b are numbers such that $f(a)f(b) < 0$, there exists a number r between a and b which is a root of f ;
- (4) if the degree n of f is one of the odd positive integers $1, 3, 5, \dots, 2m-1, \dots$, then there is a number r which is a root of f ; and
- (5) if z is a root of f , then $\bar{z}$, the conjugate of z , is a root of f .

Note. The polynomial f of the example preceding Exercise 5 has a root between 3 and 4.

Theorem 2. (The Translation Process) If f is a polynomial of positive degree n , r is a point and the Substitution Process is applied successively to obtain

$$\begin{aligned} f(z) &= (z-r)g(z) + A_n, \\ g(z) &= (z-r)g_1(z) + A_{n-1}, \\ g_1(z) &= (z-r)g_2(z) + A_{n-2}, \\ &\dots \\ g_{n-2}(z) &= (z-r)g_{n-1}(z) + A_1, \text{ and} \\ g_{n-1}(z) &= A_0. \end{aligned}$$

where g is a polynomial of degree $n-1$, g_1 is a polynomial of degree $n-2, \dots$, g_{n-1} a polynomial of degree 0 and each of $A_0, A_1, \dots, A_n$ is a point then, for every point z ,

$$f(z+r) = \sum_{p=0}^n A_p z^{n-p}.$$

The actual computation may be carried out by simply extending the table used in the substitution process as illustrated by the following example.

Let $f(z) = z^3 - 3z^2 + 2z - 7$ and $r = 1$:

$$\begin{array}{l} \begin{array}{|c|c|c|c|} \hline 1 & -3 & 2 & -7 \\ \hline & 1 & -2 & 0 \\ \hline \end{array} 1 \\ \begin{array}{|c|c|c|} \hline 1 & -2 & 0 \\ \hline & 1 & -1 \\ \hline \end{array} -7 = A_3 \\ \begin{array}{|c|c|} \hline 1 & -1 \\ \hline & 1 \\ \hline \end{array} -1 = A_2 \\ \begin{array}{|c|} \hline 1 \\ \hline \end{array} 0 = A_1 \\ 1 = A_0 \end{array}$$

$$f(z+1) = z^3 - z - 7, g(z) = z^2 - 2z, g_1(z) = z - 1, \text{ and } g_2(z) = 1.$$

We have seen that the polynomial of this example has a root between 3 and 4 (See the table preceding Exercise 5.) The translation process may be used to compute this root to any desired number of digits. Let the root be denoted by r . Then, $r = 3 + d$, where $0 < d < 1$. By means of the translation process, we now compute $f(3+d)$:

$$\begin{array}{c}
\begin{array}{|c|c|c|c|} \hline 1 & -3 & 2 & -7 \\ \hline & 3 & 0 & 6 \\ \hline \end{array} 3 \\
\begin{array}{|c|c|c|} \hline 1 & 0 & 2 \\ \hline & 3 & 9 \\ \hline \end{array} -1 \\
\begin{array}{|c|c|} \hline 1 & 3 \\ \hline & 3 \\ \hline \end{array} 11 \\
\begin{array}{|c|} \hline 1 \\ \hline \end{array} 6 \\
1
\end{array}$$

$f(3+d) = d^3 + 6d^2 + 11d - 1$. This is 0, since $r = d + 3$. I. e.,
 $= d^3 + 6d^2 + 11d - 1 = 0$ or

$$d = \frac{1}{11} - E, \text{ where } E = \frac{d^3 + 6d^2}{11}.$$

Since $0 < d < 1$, $E > 0$, and $0 < d < \frac{1}{11}$ it follows that $0 < E < 0.0045762$,
 $0.86 < d < 0.091$, and $3.086 < r < 3.091$.

We now let $r = 3.086 + d'$, $0 < d' < 0.005$ and compute
 $f(3.086 + d') = f(3 + .086 + d')$:

$$\begin{array}{c}
\begin{array}{|c|c|c|c|} \hline 1 & 6 & 11 & -1 \\ \hline & 0.086 & 0.523396 & 0.991012056 \\ \hline \end{array} 0.086 \\
\begin{array}{|c|c|c|} \hline 1 & 6.086 & 11.523396 \\ \hline & 0.086 & 0.530792 \\ \hline \end{array} -0.008987944 \\
\begin{array}{|c|c|} \hline 1 & 6.172 \\ \hline & 0.086 \\ \hline \end{array} 12.054188 \\
\begin{array}{|c|} \hline 1 \\ \hline \end{array} 6.258 \\
1
\end{array}$$

$(d')^3 + 6.258(d')^2 + 12.054188d' - 0.00898744 = 0$ Consequently, $d' =$
 $0.0007456283 - E'$, where

$$E' = \frac{(d')^3 + 6.258(d')^2}{12.054188}.$$

Since $E' > 0$, $d' < 0.0008$ and therefore $E' < 0.0000003323$ and
 $0.0007452960 < d' < 0.0007456283$, $3.0867452960 < r < 3.0867456283$.

If $r = 3.0867452 + d''$, then $0 < d'' < 0.0000005$. At the next step we
compute $f(3.0867452 + d'') = f(3.086 + 0.0007452 + d'')$:

$$\begin{array}{c}
\begin{array}{|c|c|c|c|} \hline 1 & 6.258 & 12.054188 & -0.00987944 \\ \hline & 0.0007452 & 0.00466401692304 & 0.00898625 \\ \hline \end{array} 0.0007452 \\
\begin{array}{|c|c|c|} \hline 1 & 6.2587452 & 12.05885201692304 \\ \hline & 0.0007452 & 0.00466457224608 \\ \hline \end{array} -0.00000168774... \\
\begin{array}{|c|c|} \hline 1 & 6.2594904 \\ \hline & 0.0007425 \\ \hline \end{array} 12.063516... \\
\begin{array}{|c|} \hline 1 \\ \hline \end{array} 6.2602356 \\
1
\end{array}$$

where $d'' = 0.000\,000\,139\,882\,676\ldots$ and $E'' = \frac{(d'')^3 + 6.2602356(d'')^2}{12.063516\ldots}$.

Thus $d'' < 0.000\,000\,2$ and $E'' < 0.000\,000\,000\,000\,0208$,
 $0.000\,000\,139\,882\,655\,7 < d'' < 0.000\,000\,139\,882\,676\,5$
 so that $\underline{3.086\,745\,339\,882\,655\,7} < r < \underline{3.086\,745\,339\,882\,676\,5}$
 or $r = 3.086\,745\,339\,882\,65 + d''$, $0 < d'' < 0.000\,000\,000\,000\,003$.

We have obtained :

- (1) $r = 3.09$, to two decimal places at the first step;
- (2) $r = 3.08675$, to five decimal places at the second step, and;
- (3) $r = 3.086\,745\,339\,882\,7$, to thirteen decimal places at the third step.

Exercise 6. Let f be the polynomial defined by

$$f(x) = x^4 + 2x^3 - 22x^2 - 29x + 76.$$

Show that f has a root p between -5 and -4, a root q between -3 and -2, a root r between 1 and 2, and a root s between 4 and 5. Show that $s = 4.0525621$, to seven decimal places. Find the root r correct to seven decimal places.

Exercise 7. Show that there exists only one positive number x such that $x^3 = 2$. Compute x correct to seven decimal places.

Theorem 3. (The Highest Common Factor Process) Suppose that f_0 is a polynomial of degree n_0 , f_1 a polynomial of degree n_1 , $n_0 \geq n_1$,

$$f_0(z) = \sum_{p=0}^{n_0} a_p z^{n_0-p} \text{ and } f_1(z) = \sum_{p=0}^{n_1} b_p z^{n_1-p}.$$

Then $f_0(z) = \frac{a_0}{b_0} z^{n_0-n_1} f_1(z) + g(z)$ where $g(z) = 0$ for every point z or else g is a polynomial of degree less than n_0 :

$$g(z) = \sum_{p=0}^{n_0-1} \frac{b_0 a_{p+1} - a_0 b_{p+1}}{b_0} z^{n_0-1-p}.$$

There exists a polynomial q_1 such that

$$f_0(z) = q_1(z)f_1(z) - f_2(z),$$

where $f_2(z) = 0$ for every point z , or else f_2 is a polynomial of degree $n_2 < n_1$. In the latter event, there exists a polynomial q_2 such that

$$f_1(z) = q_2(z)f_2(z) - f_3(z)$$

where $f_3(z) = 0$ for every point z , or else f_3 is a polynomial of degree $n_3 < n_2$. In the latter event, there exists a polynomial q_3 such that

$$f_2(z) = q_3(z)f_3(z) - f_4(z),$$

where $f_4(z) = 0$ for every point z , or else f_4 is a polynomial of degree $n_4 < n_3$.

There exists a positive integer m and polynomials $q_1, q_2, \dots, q_m$ such that

$$f_{p-1} = q_p(z)f_p(z) - f_{p+1}(z), p = 1, 2, \dots, m$$

where $f_{m+1}(z) = 0$ for every point z and, if $m > 1$, f_p is a polynomial of degree $n_p < n_{p-1}$, $p = 2, 3, \dots, m-1$. The polynomial f_m is a factor of both f_0 and f_1 , and any polynomial which is a factor of both f_0 and f_1 is a factor of f_m .

The process for computing $f_2, \dots, f_m, q_1, \dots, q_m$ may be conveniently arranged in a tabular form as illustrated by the following example.

Example 1. $f_0(x) = x^3 + 7x^2 + 9x - 1$, $f_1(x) = x^2 + 5x - 1$;

$$\begin{array}{r}
 1 + 2 \\
 1 + 5 - 1 \overline{) 1 + 7 + 9 - 1} \\
 \underline{1 + 5 - 1} \\
 2 + 10 - 1 \\
 \underline{2 + 10 - 2} \\
 1
 \end{array}$$

Thus, $q_1(x) = x + 2$, $-f_2(x) = 1$ and $f_2(x) = -1$.

Continuing,

$$\begin{array}{r}
 -1 - 5 + 1 \\
 -1 \overline{) 1 + 5 - 1} \\
 \underline{1} \\
 5 \\
 \underline{5} \\
 -1 \\
 \underline{-1} \\
 0.
 \end{array}$$

Thus, $q_2(x) = -x^2 - 5x + 1$. In this case, $m = 2$. No polynomial of positive degree is a common factor of f_0 and f_1 .

2. Sturm's Series. The statement that $f_0, \dots, f_m$ is a Sturm's Series means m is a positive integer, f_p is a polynomial, each of whose coefficients is a number, $p = 0, \dots, m$, $f_m(x) \neq 0$ for every number x and, if p is a positive integer less than m and x a number such that $f_p(x) = 0$, then $f_{p-1}(x)f_{p+1}(x) < 0$.

Example. If each coefficient in each of the polynomials f_0 and f_1 of Theorem 3 is a number, and $f_m(x) \neq 0$ for every number x , then $f_0, \dots, f_m$ is a Sturm's series.

If $f_0(x) = x^4 - 7x^3 - 7x^2 + 43x + 42$, $f_1(x) = -7x^3 + 43x$, $f_2(x) = x^2 - 49$, $f_3(x) = x$, and $f_4(x) = 1$, then

$$\begin{aligned}
 f_0(x) &= \left(1 - \frac{1}{7}x\right) f_1(x) - \frac{6}{7}f_2(x), \\
 f_1(x) &= -8xf_2(x) - 300f_3(x), \\
 f_2(x) &= xf_3(x) - 49f_4(x), \\
 f_3(x) &= xf_4(x), \text{ and} \\
 f_0, \dots, f_4 &\text{ is a Sturm's series.}
 \end{aligned}$$

For each number x , let $V(x)$ denote the number of 'variations of signs' in the Sturm's series $f_0, \dots, f_m$, zero terms having been dropped out before the number

of variations of signs is counted. The meaning of $V(x)$ may perhaps be best made clear by the study of the table.

x	signs	$V(x)$
-4	+ + - - +	2
-2	0 - - - +	1
0	+0 - 0 +	2
2	+ + - + +	2
3	0 - - + +	1
5	- - - + +	1
7	0 - 0 + +	1
8	+ - + + +	2

The Sturm's series used is $f_0, \dots, f_4$ defined above.

There exists an integer h , and an integer k and a positive number N such that $V(x) = h$ if $x < -N$, $V(x) = k$ if $x > N$. We denote h and k by $V(-\infty)$ and $V(+\infty)$ respectively. In the example above $V(-\infty) = 2$ and $V(+\infty) = 2$.

Suppose r is a number such that $f_0(r) = 0$. By the definition of a Sturm's series, $f_1(r) \neq 0$. There exists a positive number δ such that the quotient

$$\frac{f_1(x)}{f_0(x)}$$

is

- (i) positive if $r - \delta < x < r$ and positive if $r < x < r + \delta$, or
- (ii) negative if $r - \delta < x < r$ and negative if $r < x < r + \delta$, or
- (iii) positive if $r - \delta < x < r$ and negative if $r < x < r + \delta$, or
- (iv) negative if $r - \delta < x < r$ and positive if $r < x < r + \delta$.

The index of the quotient f_1/f_0 at r is :

$$T(r) = \begin{cases} 0, & \text{if (i) or (ii) is true,;} \\ 1, & \text{if (iv) is true, and;} \\ -1, & \text{if (iii) is true.} \end{cases}$$

If a and b are numbers, neither of which is a root of f_0 , and $a < b$, the index of the quotient f_1/f_0 from a to b , denoted by

$$I_a^b(f_1/f_0)$$

is 0 if f_0 has no root between a and b and is $\Sigma T(r)$ if f_0 has a root between a and b , the sum Σ being taken for each root of f_0 between a and b .

Theorem 4. If $f_0, \dots, f_m$ is a Sturm's series and a and b are numbers, neither of which is a root of f_0 , and $a < b$, then

$$V(a) - V(b) = I_a^b(f_1/f_0).$$

CHAPTER IV

LOCATION OF ROOTS

If n is a positive integer and $z_1, \dots, z_n$ are any n points whatsoever, there is a polynomial whose roots are $z_1, \dots, z_n$, namely the polynomial obtained by multiplying together the factors $(z - z_1)(z - z_2) \cdots (z - z_n)$. If we denote this polynomial by f , then

$$f(z) = z^n + a_1 z^{n-1} + \cdots + a_n,$$

where

$$a_1 = -(z_1 + z_2 + \cdots + z_n),$$

$$a_2 = z_1 z_2 + \cdots + z_1 z_n + z_2 z_3 + \cdots + z_2 z_n + \cdots + z_{n-1} z_n$$

the sum of all possible products of the roots taken two at a time,

$$a_3 = -(z_1 z_2 z_3 + z_1 z_2 z_4 + \cdots),$$

the negative of the sum of all possible products of the roots taken three at a time, and so on. The problem of the present chapter is to obtain information concerning the roots of any polynomial f whose coefficients are given. Since the roots do not depend essentially upon the coefficient of the highest power in f , we shall assume that this coefficient, a_0 , is 1. Otherwise, we could consider instead of f the polynomial $(1/a_0) \cdot f$.

1. The first test-sequence for a polynomial. Suppose f is the polynomial of positive degree defined by

$$f(z) = z^n + a_1 z^{n-1} + \cdots + a_n.$$

If x is a number, $f(x)$ is the point

$$(x^n + \operatorname{Re} a_1 x^{n-1} + \cdots + \operatorname{Re} a_n, \operatorname{Im} a_1 x^{n-1} + \cdots + \operatorname{Im} a_n)$$

which may be written:

$$f(x) = f_0(x) + i f_1(x),$$

where i is the point $(0, 1)$,

$$f_0(x) = x^n + (\operatorname{Re} a_1)x^{n-1} + \cdots + (\operatorname{Re} a_n)$$

and

$$f_1(x) = (\operatorname{Im} a_1)x^{n-1} + (\operatorname{Im} a_2)x^{n-2} + \cdots + (\operatorname{Im} a_n).$$

The number t is a root of f only if $f_0(t) = 0$ and $f_1(t) = 0$ since only then is it true that $f(t) = (0, 0)$.

Definition. If for $i = 1, 2, \dots, n$, $\operatorname{Im} a_i = 0$, then the first test sequence for f consists of the single term

$$f_0;$$

If $\text{Im } a_i \neq 0$ for some positive integer i not greater than n , and f_1 is a factor of f_0 , then the first test sequence consists of the two terms

$$f_0, f_1;$$

and, if $\text{Im } a_i \neq 0$ for some positive integer i not greater than n and f_1 is not a factor of f_0 , then the first test sequence for f consists of at least three terms

$$f_0, f_1, \dots, f_m$$

connected by the relations of the highest common factor process :

$$f_{p-1}(x) = q_p(x)f_p(x) - f_{p+1}(x), p = 1, \dots, m,$$

where q_p is a polynomial, $f_{m+1}(x) = 0$ for every number x and $f_2, \dots, f_m$ are polynomials and the degree of f_p is less than the degree of f_{p-1} , for $p = 2, \dots, m$.

Theorem 1. The multiplicity sum for the polynomial f in the X-axis is the same as the multiplicity sum for f_m in the X-axis, where f_m is the last term of the first test-sequence for f . Moreover, any root of f_m is a root of f .

Note. By Theorem 1, the problem of finding the roots of a polynomial which are on the X-axis, i.e., the numbers, is reduced to the problem of finding such roots for the polynomial f_m each of whose coefficients is a number. In Chapter III we have shown by examples one way to approach the latter problem.

2. The second test-sequence for a polynomial Let f be the polynomial of positive degree n defined by :

$$f(z) = z^n + a_1 z^{n-1} + \dots + a_n.$$

Definition. The second test-sequence for the polynomial f is the first test-sequence for the polynomial g defined by :

$$g(z) = i^n f(-iz),$$

where $i = (0, 1)$, and is denoted by

$$g_0, \dots, g_k.$$

Exercise 1. Show that

$$g_0(x) = x^n - (\text{Im } a_1)x^{n-1} - (\text{Re } a_2)x^{n-2} + (\text{Im } a_3)x^{n-3} + (\text{Im } a_4)x^{n-4} - \dots$$

and, if $k \geq 1$,

$$g_1(x) = (\text{Re } a_1)x^{n-1} - (\text{Im } a_2)x^{n-2} - (\text{Re } a_3)x^{n-3} + (\text{Im } a_4)x^{n-4} + (\text{Re } a_5)x^{n-5} - \dots$$

Theorem 2. The multiplicity sum for the polynomial f in the Y-axis is the multiplicity sum in the X-axis of the last term g_k in the second test-sequence for f . In order for $-ix$ to be a root of f in the Y-axis of multiplicity h it is necessary and

sufficient that x should be a root in the X-axis of g_k of multiplicity h .

Note. By Theorem 2, the problem of finding a root of f in the Y-axis is reduced to the problem of finding a root in the X-axis of a polynomial each of whose coefficients is a number.

3. Multiplicity sums in the upper and lower half-planes. The upper half-plane is the set of all points with positive ordinates and the lower half-plane is the set of all points with negative ordinates. Let f be the polynomial of positive degree n defined by:

$$f(z) = z^n + a_1 z^{n-1} + \cdots + a_n.$$

We denote by U , r and L the multiplicity sums for f in the upper half-plane, the X-axis, and the lower half-plane, respectively.

We suppose for the present that $r = 0$.

If $z_1, z_2, \dots, z_h$ are the roots of f , of multiplicities $m_1, m_2, \dots, m_h$ respectively, and x is a number, then

$$\begin{aligned} f(x) &= (x - z_1)^{m_1} (x - z_2)^{m_2} \cdots (x - z_h)^{m_h} \\ &= f_0(x) + i f_1(x), \end{aligned}$$

where f_0 and f_1 are the first two terms of the first test-sequence for f , or $f = f_0$ and $f_1(x) = 0$ for every number x . Since, as we have assumed for the present, $r = 0$, $(x - z_p) \neq 0$ for $p = 1, 2, \dots, h$.

Consider the line segment with one end x , i.e., $(x, 0)$, and the other end z_p . (See the figure.) The line $L(z_p, x)$ is parallel to the line $L(0, x - z_p)$. There exists a number θ_p between 0 and 2 such that

$$\begin{aligned} x - z_p &= |x - z_p| (\cos(\pi\theta_p) + i \sin(\pi\theta_p)), \\ (x - z_p)^{m_p} &= |x - z_p|^{m_p} (\cos(\pi m_p \theta_p) + i \sin(\pi m_p \theta_p)) \end{aligned}$$

and therefore

$$f(x) = |f(x)| (\cos(\pi\theta) + i \sin(\pi\theta))$$

where $\theta = m_1\theta_1 + m_2\theta_2 + \cdots + m_h\theta_h$. Since $f(x) = f_0(x) + i f_1(x)$,

$$f_0(x) = |f(x)| \cos(\pi\theta) \quad \text{and} \quad f_1(x) = |f(x)| \sin(\pi\theta).$$

If we write $\theta(x)$ instead of θ to indicate dependence upon x :

$$\boxed{\tan(\theta(x)) = \frac{f_1(x)}{f_0(x)}}$$

Figure 2 shows the geometry upon which the following remark is based.

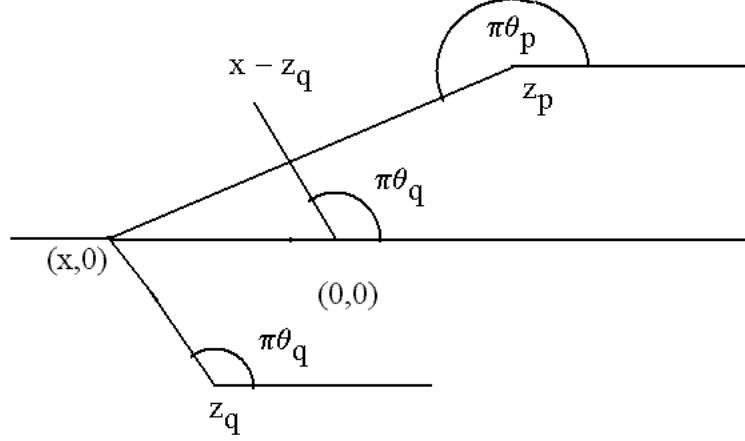


FIGURE 2. Graph Showing the Angles Determined by $(x, 0)$, z_q and z_p , with all Angles Measured Counter-Clockwise

If c is a positive number, there exists a positive number M such that if $x \leq -M$, $\theta(x)$ differs from the degree n of f by less than c . We abbreviate this statement as follows :

$$\theta(-\infty) = n.$$

There exists a positive number k such that if $x > k$, $\theta(x)$ differs from $n - L + U$ by less than c . We abbreviate this as follows :

$$\theta(+\infty) = n - L + U.$$

Thus, $\theta(-\infty) - \theta(+\infty) = L - U$. We denote this number by Δ . Then,

$$\Delta = L - U$$

$$n = L + U$$

and

$$\boxed{L = \frac{n + \Delta}{2}, U = \frac{n - \Delta}{2}.}$$

If $r > 0$, f , f_0 , f_1 have a common factor which is a polynomial of degree r . If this factor is removed, the degree of the resulting three polynomials are decreased by r , while $\tan(\pi\theta(x))$ is unchanged. On applying the preceding considerations to the polynomial F of degree $n - r$ obtained from f by removal of this factor, we obtain the formulas :

$$L = \frac{n-r+\Delta}{2}, U = \frac{n-r-\Delta}{2}.$$

The number Δ , and therefore U and L , can be obtained from a graph of the equation

$$y = \frac{f_1(x)}{f_0(x)} = \tan(\pi\theta(x))$$

because, from this graph we can determine the net decrease $\theta(-\infty) - \theta(+\infty)$ in $\theta(x)$.

Example. Suppose

$$f(z) = z^4 + (2-i)z^3 + 5iz^2 + 6z - i.$$

Then,

$$\tan(\pi\theta(x)) = \frac{-x^3 + 5x^2 - 1}{x^4 + 2x^3 + 6x}.$$

To sketch the graph of $y = \frac{-x^3 + 5x^2 - 1}{x^4 + 2x^3 + 6x}$, we sketch the graphs of the two equations

$$y = -x^3 + 5x^2 - 1$$

and

$$y = x^4 + 2x^3 + 6x$$

and then obtain the required graph by dividing the corresponding ordinates.

Remark 1. The sketch is shown on **Figure 3**. **Figure 4** shows the graph of $y = \tan(\pi\theta(x)) = \frac{-x^3 + 5x^2 - 1}{x^4 + 2x^3 + 6x}$. Consider the graph to consist of three branches, one on the right of the X -axis, one between the X -axis and the vertical line $x = -2.77766\cdots$, and the third to the left of the vertical line $x = -2.77766\cdots$. Determination of the angle $\pi\theta(x)$ in the rightmost branch is straight-forward; if $x > 0$ and $y > 0$, $\pi\theta(x)$ is in Quadrant I and if $x > 0$ and $y < 0$, $\pi\theta(x)$ is in Quadrant IV.

As π is the period of the tangent function, the angles associated with $\pi\theta(x)$ in the middle branch are π more than the angles in the right branch. Accordingly, Quadrant I angles in the right branch are associated with Quadrant III angles in the middle branch and Quadrant IV angles in the right branch are associated with Quadrant II angles in the middle branch. As the angles associated with $\pi\theta(x)$ in the left-most quadrant are 2π { i.e., 2 periods of the tangent function } more than the angles associated with $\pi\theta(x)$ in the right-most branch, Quadrant I angles in the right-most branch are associated with Quadrant I angles in the left-most branch.

Figure 5, the graph of $y = \arctan\left(\frac{-x^3 + 5x^2 - 1}{x^4 + 2x^3 + 6x}\right)$, consists of three branches, each of amplitude π . To construct a single graph, we lower the points of the middle branch by π and the left-most branch by 2π . Note that the ordinates of points of the left-most branch are contained in $[0, \pi/2]$, requiring that ordinates of points of the graph in **Figure 5** be contained in $[-2\pi, \pi/2]$. Finally, we create **Figure 6** by dividing the ordinates by π to create the graph of $y = \theta(x)$ whose ordinates are contained in $[-2, \frac{1}{2}]$.

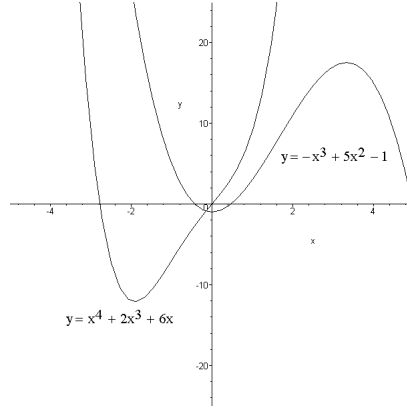


FIGURE 3. Graphs of $y = -x^3 + 5x^2 - 1$ and $y = x^4 + 2x^3 + 6x$

The roman numerals indicate the quadrants of the angle $\pi\theta(x)$.

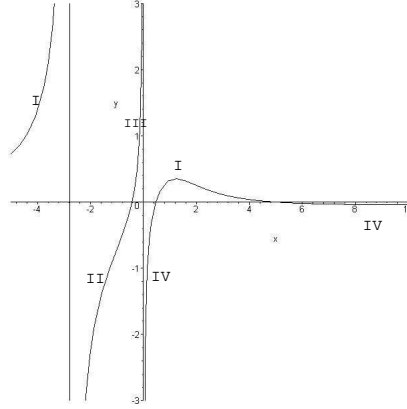
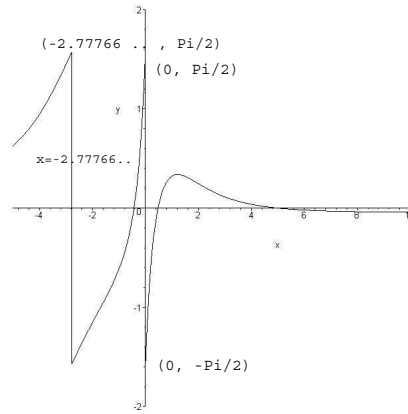
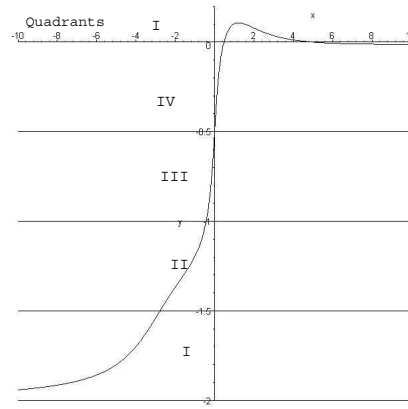


FIGURE 4. Graph of $y = \left(\frac{-x^3 + 5x^2 - 1}{x^4 + 2x^3 + 6x} \right)$ with Quadrants Annotated

We see that $\Delta = \theta(-\infty) - \theta(+\infty) = -2$ and, with $r = 0$:

$$L = \frac{4 - 2}{2} = 1 \quad , \quad U = \frac{4 + 2}{2} = 3.$$

Theorem 3. Suppose f is the polynomial of positive degree n , defined by $f(z) = z^n + a_1z^{n-1} + \cdots + a_n$, and for each number x , $V_f(x)$ is the number of variations in signs in the first test-sequence $f_0, \cdots, f_m(x)$ for f . If $\Delta = V_f(-\infty) - V_f(+\infty)$, r is the multiplicity sum for f in the X-axis, U the multiplicity sum for f in the upper

FIGURE 5. Graph of $\arctan \left(\frac{-x^3 + 5x^2 - 1}{x^4 + 2x^3 + 6x} \right)$ FIGURE 6. Graph of $y = \theta(x)$

half-plane, and L the multiplicity sum for f in the lower half plane, then

$$L = \frac{n - r + \Delta}{2}, \quad U = \frac{n - r - \Delta}{2}.$$

Note. If each coefficient of f is a number, the first test-sequence for f has only one term, $f_0 = f$, $\Delta = 0$ and $U = L$.

Example. For the polynomial $f(z) = z^4 + (2 - i)z^3 + 5iz^2 + 6z - i$ the first test-sequence is :

$$f_0(x) = x^4 + 2x^3 + 6x,$$

$$f_1(x) = -x^3 + 5x^2 - 1,$$

$$f_2(x) = -35x^2 - 5x + 7,$$

$$f_3(x) = \frac{229}{245}x - \frac{1}{35},$$

$$f_4(x) = (\text{a negative number}).$$

Hence, $V_f(-\infty)$ is the number of variations in $++--$, or 1; and $V_f(+\infty)$ is the number of variations in signs $+-+-$, or 3. Thus, $\Delta = 1 - 3 = -2 = \theta(-\infty) - \theta(+\infty)$, which agrees with the result found graphically.

4. Multiplicity sums in the right and left half-planes. The right half-plane is the set of points with positive abscissas; the left half-plane is the set of all points with negative abscissas. We denote by D and S the multiplicity sum for the polynomial f in the right half-plane and left half-plane, respectively.

The numbers U and L for the polynomial g defined by $g(z) = i^n f(-iz)$ are the numbers D and S, respectively, for the polynomial f.

Theorem 4. If f is the polynomial of positive degree n defined by $f(z) = z^n + a_1 z^{n-1} + \cdots + a_n$, s is the multiplicity sum for f in the Y-axis, D is the multiplicity sum for f in the right half-plane, S is the multiplicity sum for f in the left half-plane, $V_g(x)$ is the number of variations in signs in the second test-sequence $g_0(x), g_1, \cdots, g_k(x)$ and $G = V_g(-\infty) - V_g(+\infty)$, then

$$D = \frac{n - s - G}{2}, \quad S = \frac{n - s + G}{2}.$$

In certain problems it is important to have an easy method for determining whether or not each root of a polynomial is in the left half-plane, i.e., whether or not $S = n$, the degree of the polynomial.

Theorem 5. Suppose f is the polynomial of positive degree n defined by $f(z) = z^n + a_1 z^{n-1} + \cdots + a_n$. In order for each root of f to be in the left half-plane it is necessary and sufficient that the second test-sequence for f should contain $n+1$ terms $g_0, g_1, \cdots, g_n$ connected by the relations

$$g_{p-1}(x) = (c_p x + k_p)g_p(x) - g_{p+1}(x), \text{ for } p = 1, 2, \dots, n$$

where

$$g_{n+1}(z) = 0 \text{ and } c_p > 0 \text{ for } p = 1, 2, \dots, n.$$

Moreover, if the second test-sequence for f has $n+1$ terms and u of the terms of the sequence $c_1, c_2, \cdots, c_n$ are positive and $n-u$ are negative, then $S = u$ and $D = n - u$.

Example. Suppose $f(z) = z^5 + 2z^4 + 2z^3 + 3z^2 + z + 4$. Then, $g_0(x) = x^5 - 2x^3 + x$, $g_1(x) = 2x^4 - 3x^2 + 4$ and

$$2 + 0 - 3 + 0 + 4 \frac{\frac{1/2}{1+0-2+0+1+0}}{\frac{1+0-\frac{3}{2}+0+2}{-(\frac{1}{2}+0+1+0)}} \frac{4}{2+0-3+0+4} \frac{\frac{2+0+4+0}{-(7+0-4)}}{\frac{1/14}{\frac{1}{2}+0+1+0}} \frac{\frac{1}{2}+0-\frac{2}{7}}{-(-\frac{9}{7})} - \frac{\frac{49}{9}}{7+0-4} \frac{-\frac{9}{28}}{-\frac{9}{7}+0} \frac{7+0}{-(4)} \frac{-\frac{9}{7}}{0}$$

so that $c_1 = \frac{1}{2}$, $c_2 = 4$, $c_3 = \frac{1}{14}$, $c_4 = -\frac{49}{9}$ and $c_5 = -\frac{9}{28}$. Thus, $S = 3$ and $D = 2$. The other terms of the second test-sequence are $g_2 = \frac{1}{2}x^3 + x$, $g_3(x) = 7x^2 - 4$, $g_4(x) = -\frac{9}{7}x$, and $g_5(x) = 4$: $V_g(-\infty) = 3$, $V_g(+\infty) = 2$, $G = 1$, $D = \frac{5-1}{2} = 3$.

5. Multiplicity sum in each of the four quadrants. The first quadrant is the common part of the upper half-plane and the right half-plane; the second quadrant is the common part of the upper half-plane and the left half-plane; the third quadrant is the common part of the lower half-plane and the left half-plane; and the fourth quadrant is the common part of the lower half-plane and the right half-plane. We denote by N_1, N_2, N_3 , and N_4 the multiplicity sum for f in the first quadrant, second quadrant, third quadrant, and fourth quadrant, respectively.

We assume that f has no root on the X-axis and no root on the Y-axis, i.e.,

$$r = 0 \quad \text{and} \quad s = 0.$$

Then,

$$N_2 = U - N_1, \quad N_3 = S - N_2 = S - U + N_1 \quad \text{and} \quad N_4 = D - N_1.$$

If each coefficient of f is real, then $N_1 = D/2$. If $n = 9$, $U = 6$, $L = 3$, $D = 4$, and $S = 5$, there are four possibilities for N_1, N_2, N_3 , and N_4 :

N_1	N_2	N_3	N_4
1	5	0	3
2	4	1	2
3	3	3	1
4	2	3	0

We need a way to discover N_1 .

If the first test-sequence for f has only one term, then $N_1 = D/2$; if the second test-sequence has f for only one term, then $N_1 = U/2$. These statements must be appropriately modified in case $r \neq 0$ or $s \neq 0$.

Theorem 6. Suppose f is a polynomial of degree n defined by :

$$f(z) = z^n + a_1 z^{n-1} + \cdots + a_n,$$

with $r = 0$ and $s = 0$, the first test-sequence for f has at least two terms $0, f_1, \dots, f_k$ and the second test-sequence for f has at least two terms $g_0, g_1, \dots, g_k$. For each

number x distinct from 0, suppose

$$\phi(x) = \begin{cases} \frac{g_1(x)}{g_0(x)}, & \text{if } x < 0; \\ \frac{f_1(x)}{f_0(x)}, & \text{if } x > 0. \end{cases}$$

$T_0 = 0$ if $g_0 \neq 0$ or $f_0(0)$ and T_0 is the index of ϕ at 0 if $g_0(0) = 0$ and $f_0(0) = 0$. Let $V_g(0-)$ denote the number p such that, for some positive number c , $V_g(x) = p$ for every number x between $-c$ and 0 and $V_f(0+)$ the number q such that, for some positive number d , $V_f(x) = q$ for every number x between 0 and d . If

$$\delta = V_g(-\infty) - V_g(0) + V_f(0) - V_f(+\infty) + T_0$$

and λ is one of the numbers -1, 0, or 1, then

$$N_1 = \frac{n - 2\delta + \lambda}{4}.$$

Example. If f is the polynomial defined by $f(z) = z^4 + (1 + 2i)z^3 + (7 + i)z^2 + (9 + 5i)z - (1 - i)$, then $V_f(-\infty) = 2$, $V_f(0+) = 1$, $V_f(+\infty) = 2m$, $V_g(-\infty) = 1$, $V_g(0-) = 2$, $V_g(+\infty) = 3$, $\Delta = 0$, $G = -2$, $\delta = -2$, $U = 2$, $L = 2$, $S = 1$, $D = 3$, and $N_1 = 2$.

Exercise. Find r , U , L , s , D , S , and N_1 for

- (1) $z^3 + (1 - i)z^2 + (2 - 2i)z + (1 + i)$,
- (2) $z^4 + (2 - i)z^3 + 5iz^2 + 6z - i$,
- (3) $z^3 + (1 - i)z^2 + (2 - 2i)z + i$, and
- (4) $z^3 + (1 + 6i)z^2 - (11 - 5i)z - (7 + 9i)$.

6. A rectangle enclosing the roots. The statement that the polynomial f of positive degree n defined by $f(z) = z^n + a_1z^{n-1} + \cdots + a_n$ is regular means that there its second test-sequence contains $n + 1$ terms, $g_0, g_1, \cdots, g_n$.

Exercise 1. Show that if f is not regular, there exists a positive number h such that the polynomial F defined by $F(z) = f(z + h)$ is regular.

Example. The polynomial f defined by $f(z) = z^3 + (1 - i)z^2 + (2 - 2i)z + i$ is not regular, but the polynomial F defined by $F(z) = f(z + 1)$ is regular.

We suppose that f is regular, so that

$$g_{p-1}(z) = (c_p x + k_p)g_p(x) - g_{p+1}(x); p = 1, 2, \cdots, n, \\ g_{n+1}(x) = 0, c_p \text{ is a number distinct from 0 and } k_p \text{ is a number.}$$

Theorem 7. Let h be the polynomial defined by

$$h(z) = \frac{g_1(iz)}{c_1 i^{n-1}}$$

Then, f and h have no root in common and, if z is a point which is not a root of f :

$$\frac{h(z)}{g(z)} = \frac{1}{\frac{1-ik_1}{c_1} + z + \frac{1/(c_1c_2)}{\frac{-ik_2}{c_2} + z + \frac{(1/c_2c_3)}{\frac{-ik_3}{c^3} + z + \dots \frac{1/(c_{n-1}c_n)}{\frac{-ik_n}{c_n} + z}}$$

Thus, if $t_1, t_2, \dots, t_n$ are the linear fractional transformations defined by :

$$t_1(u) = \frac{1}{\frac{1-ik_1}{c_1} + z - u}, \quad t_p(u) = \frac{-(1/c_{p-1}c_p)}{\frac{-ik_p}{c_p} + z - u}, \quad p = 2, \dots, n.$$

then

$$\frac{h(z)}{f(z)} = t_1 t_2 \dots t_n(0)$$

Note. Compare to Theorem 25 of Chapter II.

Let a be a point with absolute value 1. We may then write :

$$\frac{h(z)}{g(z)} = \frac{\bar{a}}{\bar{a} \cdot \frac{1-ik_1}{c_1} + \bar{a}z + \frac{\bar{a}^2 \cdot (1/c_1c_2)}{\bar{a} \cdot \frac{-ik_2}{c_2} + \bar{a}z + \dots \frac{\bar{a}^2 \cdot (1/c_{n-1}c_n)}{\bar{a} \cdot \frac{-ik_n}{c_n} + \bar{a}z}}$$

We now introduce the following notation:

$$B_1(a) = \operatorname{Re} \left(\bar{a} \cdot \frac{1-ik_1}{c_1} \right), \quad B_j(a) = \operatorname{Re} \left(\bar{a} \cdot \frac{-ik_j}{c_j} \right), \quad j = 2, \dots, n,$$

$$A_j(a) = \operatorname{Im} \left(\frac{\bar{a}}{\sqrt{c_j c_{j+1}}} \right), \quad j = 1, \dots, n-1.$$

Theorem 8. Let each of $p_1, \dots, p_n$ be a positive number and suppose $m(a)$ is a number so determined that

$$B_j(a) + m(a) > p_j, \quad j = 1, \dots, n.$$

and

$$(A_{j-1}(a))^2 \leq p_{j-1} (B_j(a) + m(a) - p_j), \quad j = 2, \dots, n.$$

If z is a point such that $\operatorname{Re}(\bar{a}z) \geq m(a)$, then the quotient $\frac{h(z)}{f(z)}$ is a point, i.e., $f(z) \neq 0$.

If, for each positive integer j less than $n+1$, δ_j is a number between 0 and 1 and $m(a)$ is a number such that

$$B_j(a) + m(a) > 0 \quad , \quad j = 1, \dots, n,$$

and

$$(A_j(a))^2 \leq (B_j(a) + m(a))(B_{j+1}(a) + m(a)) \cdot \delta_j(1 - \delta_{j+1}) \quad , \quad j = 1, \dots, n-1,$$

then there exists, for each positive integer j less than $n+1$, a positive integer p_j such that

$$B_j(a) + m(a) > p_j \quad , \quad j = 1, \dots, n,$$

and

$$(A_j(a))^2 \leq p_j(B_{j+1}(a) + m(a) - p_{j+1}) \quad , \quad j = 1, \dots, n-1,$$

as is required in Theorem 8.

Theorem 9. For each of the four values $a = 1$, $a = i$, $a = -1$, and $a = -i$, let $m(a)$ be the number such that, for some conveniently chosen number δ_j , not less than 0 nor greater than 1, ($j = 1, \dots, n$), the following inequalities hold:

$$B_j + m(a) \geq 0 \quad , \quad j = 1, \dots, n,$$

and

$$(A_j(a))^2 \leq (B_j(a) + m(a))(B_{j+1}(a) + m(a)) \cdot \delta(1 - \delta_{j+1}), \quad j = 1, \dots, n-1.$$

Then, every root of the polynomial f is within or upon the rectangle with sides determined by the equations :

$$x = m(1), \quad y = m(i), \quad x = -m(-1), y = -m(i).$$

Illustrative Example. Let $f(z)$ be the polynomial defined by

$$f(z) = z^3 + (1 + 6i)z^2 - (13 - 5i)z - (7 + 10i).$$

Then,

$$g_0(x) = x^3 - 6x^2 + 13x - 10 \quad , \quad g_1(x) = x^2 - 5x + 7$$

$$1 - 5 + 7 \quad \frac{1 - 1}{\frac{1 - 6 + 13 - 10}{1 - 5 + 7}} \quad c_1 = 1, k_1 = -1$$

$$\frac{-1 + 6 - 10}{-1 + 5 - 7} \quad \frac{-1 + 2}{\frac{1 - 5 + 7}{1 - 3}} \quad c_2 = -1, k_2 = 2$$

and

$$\frac{-2 + 7}{-2 + 6} \quad \frac{1 - 3}{\frac{-1}{-1 + 3}} \quad c_3 = 1, k_3 = -3$$

$$3 \cdot \cdot \cdot$$

The values of $B_j(a)$ and $A_j(a)$ are given by the following table.

	a=1	a=i	a=-1	a = -i
$B_1(a)$	1	1	-1	-1
$B_2(a)$	0	2	0	-2
$B_3(a)$	0	3	0	-3
$(A_1(a))^2$	1	0	1	0
$(A_2(a))^2$	1	0	1	0

If we take $\delta_1 = 1$, $\delta_2 = 1/2$ and $\delta_3 = 0$, the inequalities to be satisfied by $m(a)$ are as follows : (We write m for $m(a)$.)

a=1	a = i	a = -1	a = -i
$1 + m \geq 0$	$1 + m \geq 0$	$-1 + m \geq 0$	$-1 + m \geq 0$
$m \geq 0$	$2 + m \geq 0$	$m \geq 0$	$-2 + m \geq 0$
$2 \leq (1 + m)m$	$3 + m \geq 0$	$2 \leq (-1 + m)m$	$-3 + m \geq 0$
$2 \leq m^2$	$0 \leq (1 + m)(2 + m)$	$2 \leq m^2$	$0 \leq (-1 + m)(-2 + m)$
	$0 \leq (2 + m)(3 + m)$		$0 \leq (-2 + m)((-3 + m)$

We see by inspection that these are satisfied if $m(1) = \sqrt{2}$, $m(i) = -1$, $m(-1) = 2$ and $m(-i) = 3$.

Hence, a rectangle including all of the roots of f is determined by the four straight lines whose equations are :

$$x = \sqrt{2}, \quad y = -1, \quad x = -2, \quad y = -3.$$

Exercise 2. Find the rectangle enclosing all of the roots of the polynomial f defined by

$$f(z) = z^3 + (1 - i)z^2 + (2 - 2i)z + i.$$

7. Computation of the roots. We have seen that each root of the polynomial f defined by

$$f(z) = z^3 + (1 + 6i)z^2 - (13 - 5i)z - (7 + 10i)$$

is in the rectangular region defined by $-2 \leq \operatorname{Re} z \leq \sqrt{2}$, $-3 \leq \operatorname{Im} z \leq -1$. Also, we $c_1 = 1$, $c_2 = -1$ and $c_3 = 1$, so that $D = 1$ and $S = 2$ and there is a root in the strip defined by $0 \leq \operatorname{Re} z \leq 2$.

We compute $f(z + 1)$ by the translation process :

$f(z + 1) = z^3 + (4 + 6i)z^2 - (8 - 17i)z - (18 - i) = F(z)$, and then compute the second test-sequence for F :

	.25 - .4375
4 - 17 + 18	1 - 6.00 + 8.00 + 1 1 - 4.25 + 4.5
	-1.75 + 3.50 + 1.00 -1.75 + 7.44 - 7.86
	-(3.94 - 8.86)

and, continuing

-(3.94 - 8.86)	1.015 - 2.03
	4 - 17 + 18 4 - 9
	-8 + 18 -8 + 18
	0

Thus, $g_0(x) = x^3 - 6x^2 + 8x + 1$, $g_1(x) = 4x^2 - 17x + 18$, $g_3(x) = 3.94x - 8.86$ is the second test-sequence for F. The last term has the root

$$\frac{8.86}{3.94} \approx 2.25$$

so that F has root $-2.25i$ on the Y-axis and, consequently, f has the root $1 - 2.25i$, approximately.

Let the point d be defined by $z_1 = 1 - 2.25i + d$, where z_1 is the root of f in question. We now form, by the translation process, the terms of $f(1 - 2.25i + d) = F(-2.25i - d)$ which does not involve d^3 or d^2 :

1	4 + 6i -2.25i	-8.000 + 17i +8.4375 - 9i	-18 + i 18 - 0.984375i	-2.25i
1	4 + 3.75i -2.25i	0.4375 + 8i 3.375 - 9i	0.015625i	
1	4 + 1.5i	3.8125 - i		

so that $f(1 - 2.25i + d) = d^3 + (\text{an undermined coefficient})d^2 + (3.8125 - i)d + 0.015615i = 0$.

If we neglect the terms in d^3 and d^2 , we then have, approximately,

$$d = \frac{-0.015625i}{3.8125 - i} = \frac{-0.015625i(3.8125 + i)}{3.8125^2 + 1} = 0.0010 - 0.0038i.$$

This gives for z_1 the approximation $1.0010 - 2.2528i$.

The calculation is relatively easy here because the abscissa of z_1 happens to be so near to 1.

We shall now find one of the roots in the strip defined by $-2 \leq \operatorname{Re} z \leq 0$. We first calculate $f(z - 1)$ and its second test-sequence :

$$f(z - 1) = z^3 + (-2 + 6i)z^2 + (12 - 7i)z + (6 + 9i),$$

	-0.5 + 1.25
-2 + 7 - 6	1-6.0+12.00 + 9.0 1-3.5+ 3.00
	-2.5+9.00 + 9.0 -2.5 +8.75 - 7.5
	-(0.25-16.5)

and

	8 - 556	
-(-0.25 - 16.5)	-2 + 7 -6 -2 - 132	
	139 -6 139 + 9174	(-)
	-(9180)	-0.25 - 16.5

so that $c_1 < 0$, $c_2 > 0$, $c_3 < 0$ and consequently f has a root in each of the strips $-2 \leq \operatorname{Re} z \leq -1$ and $-1 \leq \operatorname{Re} z \leq 0$. In order to find the root in the latter strip we form $f(z - 0.5)$ and its second test-sequence.

$$f(z - 0.5) = z^3 + (-0.5 + 6i)z^2 + (-13.25 - i)z + (-0.375 - 11i),$$

		-2 + 8	
-0.5 + 1 + 0.375	1 -6 +13.25 -11 1 -2 -0.75		
	4 + 14 -11 -4 + 8 +3	0.08333 + 0.0277778	
	-(-6 + 14)	-0.50000 + 1.00000 + 0.3750 -0.50000 + 1.66667	
		-0.16667 + 0.3750 -0.16667 + 0.3889	(-)
		-(0.014)	-6 + 14

so that $c_1 < 0$, $c_2 > 0$, $c_3 < 0$ and there is a root, z_2 in the strip $-0.5 < \operatorname{Re} z < 0$. Now, in the second test-sequence for $f(x - 0.5)$, $g_3(x) = 0.014$, which is 0 to one decimal place. If we regard this as 0, the $g_2(x) = -6x + 14$ is the last term of the second test-sequence. Hence, the polynomial F defined by $F(z) = f(z - 0.5)$ has the root

$$-i \cdot \frac{14}{6} = -2.3i$$

and f has the root $-0.5 - 2.3i$ (approximately). If we define the point d by $z_2 = -0.5 - 2.3i + d$ and equate to 0 the term of degree lower than the second in $f(d - 0.5 - 2.3i)$, we obtain the estimates :

$$d = 0.02 - 0.05i, \quad z_2 = -0.48 - 2.35i \quad (\text{approximately}).$$

Since the sum of the roots of f is $-1 - 6i$, the remaining root is approximately

$$z_3 = -1 - 6i - (1.00 - 2.25i) - (-0.48 - 2.35i) = -1.52 - 1.40i.$$

As a check, the product of the roots is $7 + 10i$ and the product of our approximations to the roots is $6.9886 + 10.0049i$.

If greater accuracy is desired, we form $f(d - 0.48 - 2.35i)$ as far as the terms of degree less than 2 and get $d = -0.000653 + 0.0037353i$, so that $z_2 = 0.4807 - 2.3463i$ and $z_3 = -1 - 6i - (1.0010 - 2.2538i) - (0.4807 - 2.3463i) - (0.4807 - 2.3463i) = -1.5203 - 1.3999i$. To continue this process, we would form the terms of degree less than 2 in $f(d + 1.0010 - 2.2538i)$ and $f(d - 0.4807 - 2.3463i)$.

In the initial stages of the process, interpolation may sometimes be used to advantage. To illustrate, if f is the polynomial defined by

$$f(z) = z^3 - (1 - i)z^2 + (2 - 2i)z - (1 + i)$$

we find that on computing the second test-sequence for $f(z + 1)$ that

$$c_1 > 0, \quad c_2 > 0, \quad c_3 < 0, \quad g_3(x) = -1,$$

and for $f(z + 2)$

$$c_1 > 0, \quad c_2 > 0, \quad c_3 > 0, \quad g_3(x) = 6.46$$

There is a root of f in the strip $1 < \operatorname{Re} z < 2$. If we interpolate between the values -1 and 6.46 of $g_3(x)$ for $f(z + 1)$ and $f(z + 2)$, we find that the predicted value for which g_3 should be 0 is 1.13, i.e., $g_3(x)$ for the polynomial $f(z + 1.13)$ should be 0. Thus, the interpolation predicts that the root should be in the strip $1.1 < \operatorname{Re} z < 1.2$. On computing the second test-sequence for $f(z + 1.2)$, we find that $g_3(x) = -0.024$ which is 0 to one decimal place. If we regard $g_3(x)$ as 0, we find the approximation is $1.2 + 0.73i$ for the root in question. If this root is $d + 1.2 + 0.73i$, we find by the process of the preceding example :

$$d = 0.0041 + 0.0033i,$$

so that the second approximation to the root is

$$1.2041 + 0.7333i$$

Exercise 1. Find approximates to the remaining two roots of the polynomial just considered.

Exercise 2. Show that each root of the polynomial defined by

$$f(z) = z^3 - 3z^2 + 2z - 7$$

is in the region defined by

$$-0.21 < \operatorname{Re} z < 3.21 \quad , \quad -2.2 < \operatorname{Im} z < 2.2.$$

We found in Section 1 of Chapter III that one of the roots is 3.08675, to five decimal places. Show that the remaining two roots are

$$-0.04337 + 1.50528i \quad , \quad -0.04337 - 1.50528i,$$

to five decimal places.

Exercise 3. Find the approximations to the roots of f defined by

$$f(x) = x^4 + 2x^3 + 6x.$$

MATHEMATICS DEPARTMENT, SAN ANTONIO COLLEGE, SAN ANTONIO, TEXAS 78212-4299
E-mail address: `screswell@mail.accd.edu`